3.1H AND THE START OF 3.1I · PP. 72–74

Data records part of a situation

> Identify three kinds of data in the scene. Which reason for watching would remain an inference?



AI conceptual illustration · fictional scene, not case evidence

APPLY · 2 · TEXTBOOK PP. 72–73

Big data and data analytics

Volume: amount

Velocity: arrival/processing rate

Variety: forms

Veracity: trustworthiness

A useful dataset fits the question; more records alone cannot repair missing context or systematic omission.

Big data involves large, varied collections of records

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 72–73.

Volume and velocity

Situation	Characteristic to examine
A ten-year archive	Volume
Thousands of new events each second	Velocity
A daily batch analysed overnight	Arrival and processing delay

Volume describes the amount of data

APPLY · 4 · TEXTBOOK PP. 73

Variety and veracity

VARIETY: DIFFERENT FORMS

Video + caption + device setting + interaction log

VERACITY: TRUSTWORTHINESS

Correct units? Duplicates? Missing fields? Outdated labels?

More formats and more rows do not repair the same underlying error.

Variety concerns formats and types of data

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 73.

APPLY · 5 · TEXTBOOK PP. 73–74

Small and clean, or large and messy?

A · 40 VERIFIED RECORDS

One class. Useful for that class—but does it represent the school?

B · 40,000 IMPERFECT RECORDS

Duplicates and missing fields. Wider reach—but what errors remain?

Synthetic datasets. Choose against a defined question, not size or cleanliness alone.

Dataset A: 40 verified records from one class

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 73–74.

APPLY · 6 · TEXTBOOK PP. 73–74

A missing value is an unresolved question

SAME RECORDS. DIFFERENT ASSUMPTION.

8 blank

Exclude blank: 8 ÷ 1 = 8. Assume zero: (8 + 0) ÷ 2 = 4.

The interface cannot tell you which assumption is justified. Investigate the missing record.

Blank duration does not mean zero duration

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 73–74.

APPLY · 7 · TEXTBOOK PP. 73

An association does not establish a cause

1. Observed association →

2. Alternative explanation →

3. Comparison evidence →

4. Qualified causal claim

A possible explanation requires testing. Correlation alone cannot identify intention or establish cause.

A pattern may be useful for prediction

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 73.

APPLY · 8 · TEXTBOOK PP. 72, 74

The data lifecycle creates decisions at every stage

01

Collect

Who and what is recorded?

02

Store

Who has access?

03

Use

For which purpose?

04

Retain / destroy

For how long?

A map of decision points. This assigned excerpt begins collection dilemmas; it does not teach every later lifecycle section.

Create or collect data

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 72, 74.

APPLY · 9 · TEXTBOOK PP. 74

Collection can shape the eventual result

1. Defined purpose →

2. Choose relevant fields →

3. Check omitted groups →

4. Decide retention

Collection choices shape both evidence and privacy costs.

Collect data that fits a defined purpose

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 74.

Analytics can support several purposes

DESCRIBE

What happened?

Last month’s activity

DETECT

What looks unusual?

A possible false alert

PREDICT

What may happen next?

Compare later observation with the estimate

Different questions create different errors and evidence needs.

Describe a past pattern

A bounded big-data inquiry

01

Question: are new creators represented?

→

02

Inspect collection, formats and missing records

→

03

Tie each claim to a source and a limit

A provider account and regulator account do not supply a complete dataset of individual experiences.

Question: whose experience does this dataset describe?

APPLY · 12 · TEXTBOOK PP. 72–74

Chapter coverage and delayed retrieval

PAUSE · INDIVIDUAL FIRST ATTEMPT · 2 MINUTES

Which claim is justified after this deck?

Predict / retrieve | Justify with an operation | Check → correct

Save your first response below. Explain the correction on a fresh example; practice is not a mark.

This deck teaches 3.1H and collection at the start of 3.1I

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 72–74.